Product Experience - Dim Editor Session Pipeline

Summary

This page outlines the editing session pipeline data assets required to be migrated from the existing Transform mono dbt project to the new Product Experience data domain service (HLDD).

With this migration, the Product Experience data domain is redesigning the pipelines to have a strong data warehouse foundation for any data users to contributed to, as well as shared metrics and definitions across 3 Supergroups (Design Experience, Teams and Education and Generative AI) to enable cross product, feature reporting and self-serve analytics.

Project Scope

The data and metric definition required to be migrated include:

- Dim editor session pipeline

A detailed session for the data models for each report will be listed in the following session

Current State

Dim editor session pipeline provides a dimension model `dim_editor_session` for downstream consumption, it serves dimensional attributes as well as aggregated facts related to a design session.

A design session represents an aggregation of editing events (those with an editing context and design_session_id) to represent a session of editing a design. The user's design session id (`design_session_id`) is a randomly generated string created when a design is loaded in the editor. The id was originally uuidv4, and then changed to ulid as of August 2023. It uniquely identifies continuous periods of time editing the same design on the same device without 30 mins of inactivity.

- If the user loads the same design in two tabs, that results in the same session id.
- If the user leaves a design open in a tab for more than 30 minutes, a new session id will be created.
- A transient editing session without a `design_id` (eg. anon editing) will have a different `design_session_id` once the design is assigned a `design_id`.

The editing session pipeline is made of the following models in the current Transform project:

Table Name	Purpose	Source	Contains CTEs / macros / LoC
_fact_editor_event_track_filtered	Collect events with editing_context, other attributes including page_height, page_width, network_status, view_mode and num_elements_in_page	Analytics Events	Yes / Yes / 100
_stg_editor_session_first_event	Calculate the earliest event_time of each design_session_id	_fact_editor_event_track_filtered	Yes / No / 80
_stg_editor_session_last_event	Calculate the latest event_time of each design_session_id	_fact_editor_event_track_filtered	Yes / No / 50
_stg_editor_session_details	Accumulative stats related to network, page elements, events and participant, plus indicators (is_new_design and is_remix_design) for each editing session	_fact_editor_event_track_filtered event_design_opened	Yes / No / 120
_stg_editor_session_summary	Summary of each session with editing session metadata	_stg_editor_session_first_event _stg_editor_session_last_event _dim_editor_event_track_editing_context	Yes / No / 220

_dim_editor_session	Summary of each session with aggregated stats	_stg_editor_session_summary _stg_editor_session_details _dim_mobile_performance _stg_visitor_internal dim_user	No / No / 120
dim_editor_session	View of _dim_editor_session	_dim_editor_session	No / Yes (dbt utils) / 60

Future State

dim editor session table has the following column, which is a mix of dimension and facts.

Column name	Description	Entity Type
session_date	The date the first event fired during the session i.e. session_started_at converted to a date.	Fact / degenerate dimension
session_started_at	The event time of the first event fired during the session.	Fact / degenerate dimension
session_ended_at	The event time of the last event fired during the session.	Fact / degenerate dimension
design_session_id	This field will only be populated when page = editor. The user's design session id is a randomly generated string created when a design is loaded in the editor. The id was originally uuidv4, and then changed to ulid as of August 2023. It uniquely identifies continuous periods of time editing the same design on the same device without 30 mins of inactivity.	Dimension
first_chain_session_id	The first design_session_id in the chain. Only available for logged-in users i.e. where user_id is not null.	Dimension
session_chain_started_at	The time the first design_session_id in the chain started. Only available for logged-in users i.e. where user_id is not null.	Degenerate dimension

user_id	The ID of the user, if the user was logged in or if the user was logged out but logged in on the same device and was identified using link_first_user_anonymous_id.	Dimension
anonymous_id	The unique visitor identifier generated by the Product Analytics Platform. Deprecated as of June 2023. Please use session_id or device_id. Completely NULL from 3rd of August 2023 onwards.	Dimension
session_id	Unique identifier associated with a Canva session, applies to both logged-in and logged-out user. Reset after 30 minutes of inactivity. It will only be null from the backend.	Dimension
brand_id	The ID of the user's brand.	Dimension
country_code	The 2-character country code, based on the IP address of the page view and calculated using MaxMind's geolocation databases.	Dimension
device_id	For iOS, the IDFV (identifier for vendors). For Android, this is the Android ID.	Dimension
device_name	A concatentation of device	Dimension

	model and device manufacturer.	
device_model	Phone model name, as captured by Canvalytics events.	Dimension
device_manufacturer	Manufacturer of the phone (e.g. samsung, Xiaomi).	Dimension
platform	The platform which the device was used on.	Dimension
experiments	List of experiments the user was enrolled in at the start of the session.	Dimension
browser_name	Name of the browser the event occurred on. May be null for visits within the Canva app (HomeX).	Dimension
is_anonymous_editor	Flag indicating whether the session was completed by a user in a logged-out version of the editor.	Dimension
is_design_owner	Flag indicating whether the user is the owner of the design.	Dimension
mobile_app_context	Object containing details about the Canva app build and version being loaded.	Dimension
design_id	ID of the design.	Dimension
doctype_id	Doctype ID of the design.	Dimension
category_id	The category ID of the design if applicable.	Dimension

brand_kit_id	ID of the Brand Kit (e.g. kADv-GSRqwo).	Dimension
positioning	Either 'fixed' or 'responsive', refers to whether the position of elements in the design affects other elements.	Dimension
design_owner_user_id	The user_id of the owner of the design.	Dimension
view_mode	The view mode of the editor at the start of the session.	Dimension
access_role	The level of access the user has to the design.	Dimension
network_status_start	The network status at the start of the session (online/offline).	Dimension
network_status_end	The network status at the end of the session (online/offline).	Dimension
first_event_name	The event name of the first event in the session.	Dimension
last_event_name	The event name of the last event in the session.	Dimension
device_tier	The mobile device tier where relevant. Derived from dim_hardware_classification.	Dimension
device_tier_number	The tier of the device as a number. One of 2 (low), 1 (mid), or 0 (high). It's NULL from Oct 2025.	Dimension

is_new_design	This flag can be used to determine whether the design is new or existing.	Dimension
is_remix_design	Flags whether the design is a copy of an existing design.	Dimension
analytics_correlation_id	This is an id created by a service with properties and sent to the document service to generate a design_id.	Dimension
was_design_opened	Flags whether the design was opened in this session, or this session was a continuation after a period of inactivity.	Dimension
is_internal_user	Indicates that the user is a Canva employee (based on email domain) or other test account.	Dimension
is_slow_session_measurable	Indicates whether the session supports measuring how slow the session is.	Dimension
design_complexity_tier_start	This measures the complexity of a design at the start of the session based off the volume of content and tiers.	Dimension
design_complexity_tier_end	This measures the complexity of a design at the end of the session based off the volume of content and tiers.	Dimension

session_duration_secs	The total time spent in the session between the first and last event fired.	Fact
num_pages_in_design_start	The number of pages in the design at the start of the session.	Fact
num_pages_in_design_end	The number of pages in the design at the end of the session.	Fact
num_elements_or_items_in_design_start	The number of elements/items in the design at the start of the session.	Fact
num_elements_or_items_in_design_end	The number of elements/items in the design at the end of the session.	Fact
num_characters_in_design_start	The number of characters in the whole design at the start of the session including whitespace.	Fact
num_characters_in_design_end	The number of characters in the whole design at the end of the session including whitespace.	Fact
num_quick_reopens	The number of times the design was reopened within 5 seconds of another action.	Fact
num_network_status_online	The number of events fired with a network status of online.	Fact

num_network_status_offline	The number of events fired with a network status of offline.	Fact
num_network_status_unknown	The number of events fired with a network status of unknown.	Fact
num_network_disconnections	The number of times where the current network status of an event is offline and the previous is online.	Fact
num_network_reconnections	The number of times where the current network status of an event is online and the previous is offline.	Fact
num_events	The number of events fired during the session.	Fact
max_num_elements_in_page	The max number of elements in a page in the session.	Fact
max_participant_count	The maximum participant count in the session.	Fact
design_published_count	Number of times design has been published in the session.	Fact
design_shared_count	Number of times design has been shared in the session.	Fact
session_start_logged_in_edit_users	Number of identified signed-up users in the design at the start of the session.	Fact
session_start_anonymous_edit_users	Number of unidentified users in the design at the	Fact

	start of the session.	
max_logged_in_edit_users	The maximum number of identified signed-up users in the design during the session.	Fact
max_anonymous_edit_users	The maximum number of unidentified users in the design during the session.	Fact
design_complexity_start	This measures the complexity of a design at the start of the session as a continuous number.	Fact
design_complexity_end	This measures the complexity of a design at the end of the session as a continuous number.	Fact

Expectations:

- Refactor the model to be a fact table (`fact_editor_session`), or split it into multiple models (dimension and facts) following Canva modelling standard to provide dimensional attributes and derived editing session facts for downstream uage if required
- Refactor and reduce upstream supporting models for simplicity and ease of maintenance
- New DBT model objects to be created in Product Experience Conformed service
- New DBT model objects to be created in Product Experience Metric service
- Update downstream models to point to the new DBT models

Questions:

- Who are the editor users?
- What platform do they work on?
- What's the average session duration in seconds?
- What's the total session duration in seconds by date by user?
- How many designs are users editing on a daily basis and what are the design ids?
- How many design sessions do we have per user?
- How many simultaneous collaboration editor sessions the user had during a day?